An Analysis of the COMPAS Algorithm Using Gender as the Protected Class

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Introduction

In 2016, ProPublica published an analysis of Northpointe's COMPAS algorithm, which is used to predict recidivism. Their analysis used a sample of 7,214 defendants who could be followed for two years (6,172 after ProPublica's filters) from Broward County, Florida, to compare how COMPAS scored Black and white defendants (note that ProPublica filtered the dataset for each statistical test). To assess potential bias between the subgroups, they used white defendants as their reference group for logistic regressions and Cox proportional hazards models and the Kaplan-Meier estimator to conduct survival analyses. They compared the risk classifications given to each defendant by COMPAS, scored on a scale of 1 to 10 and further grouped into Low, Medium, and High Risk categories based on the numerical score. ProPublica measured recidivism as an individual being re-arrested within the two-year period. Re-arrest does not necessarily equal reoffending, but it is used to measure recidivism throughout this analysis. They used two fairness tests, error rates and calibration, to test for bias. Error rates measure whether people are wrongly flagged at the same rate in each group. Calibration assesses whether the same score means the same thing for each group.

ProPublica found bias against Black defendants: although COMPAS largely passed calibration tests, as Northpointe argued and ProPublica's analysis also showed, it failed on error rates. Black defendants who did not recidivate were wrongly labeled higher risk 44.9% of the time vs 23.5% for white defendants, and white defendants who did recidivate were wrongly labeled low risk 47.7% vs 28.0%.

This analysis reproduces ProPublica's COMPAS analysis, using the dataset and code published on their GitHub repository, but assesses bias against female defendants instead of racial bias against Black defendants. The dataset records this variable as sex–this paper uses gender, as ProPublica did. Female gender becomes the protected class and is measured against male defendants, and race becomes one of several controls. All of ProPublica's statistical tests are reproduced, and an additional forest plot comparing hazard ratios and confidence intervals is added to this analysis.

Results

In ProPublica's sample, 64% of female defendants did not reoffend within two years compared to 53% of male defendants. In other words, a greater share of male defendants reoffended than female defendants: 36% of females reoffended compared to 47% of males.

Histograms

The histograms of the COMPAS score distributions by sex reflect the variation in count between male and female defendants: after filtering, there were 1,175 female defendants compared to 4,997 male defendants in ProPublica's sample. Female defendants' scores skewed lower: 13% of female defendants were scored with an 8–10 (highest risk) compared to 20% of male

defendants. At the low risk end, the shares were similar: 22% of females and 21% of males scored a 1 (lowest risk).

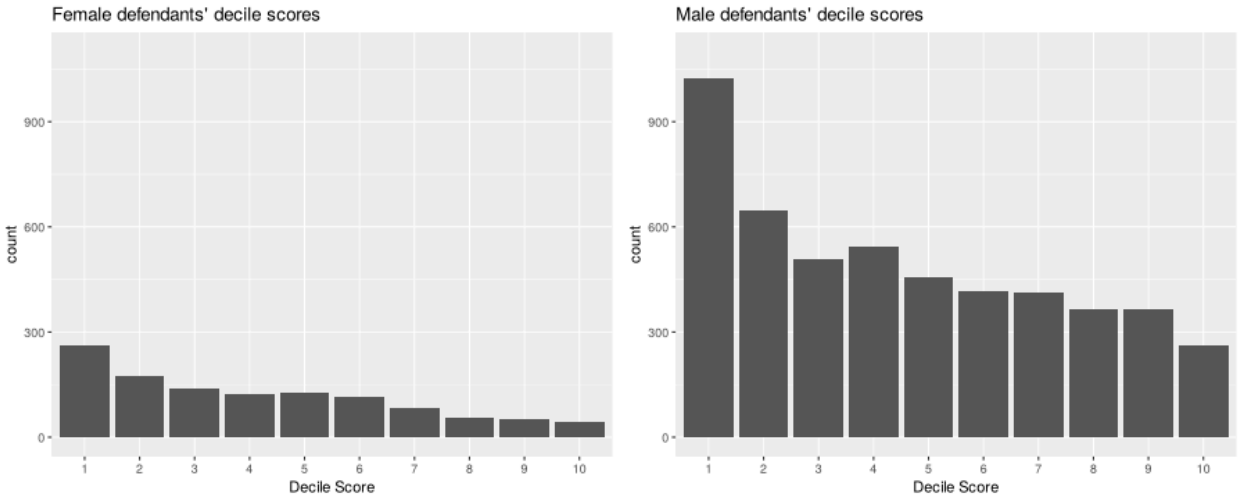


Figure 1. COMPAS decile score distributions by sex

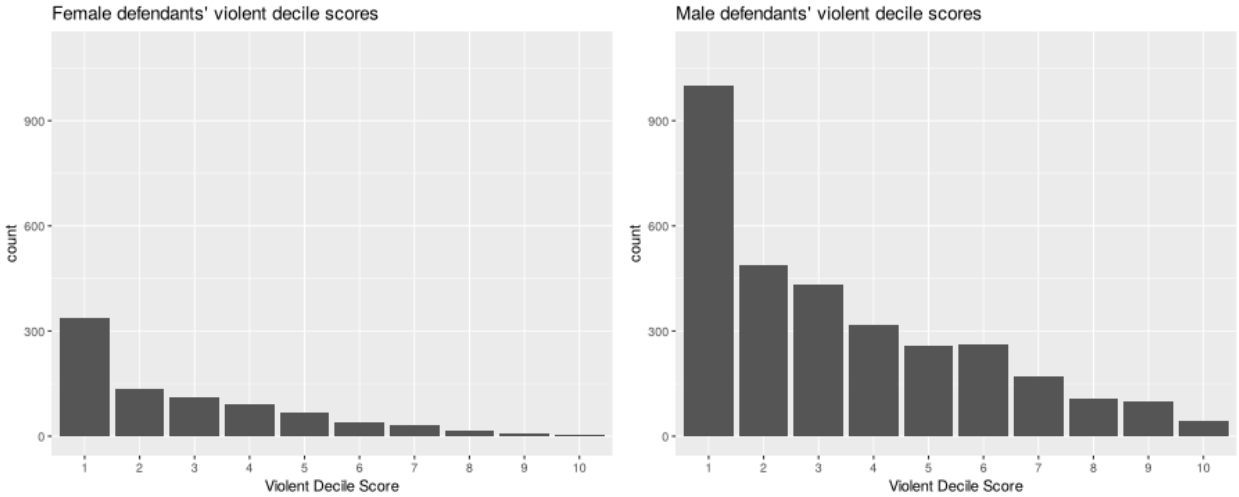


Figure 2. COMPAS violent decile score distributions by sex

Logistic Regressions

Based on the logistic regression for the overall score, in which female gender, rather than Black race, is treated as the protected group, and controlling for age, race, prior offenses, charge degree, and two-year recidivism, female defendants were 19.4% more likely than comparable male defendants to be scored as high or medium risk. The same regression on the violent score specifically showed the opposite pattern. Females were about 49% less likely than comparable males to be scored as medium or high risk on the violent score.

Error Rates

Error rates between female and male defendants were similar overall. Female defendants had a false positive rate of 32.11% (meaning 32.11% of female defendants who did not reoffend were incorrectly scored as high or medium risk) and a false negative rate of 39.16% (meaning

39.16% of female defendants who reoffended were incorrectly scored as low risk). In comparison, male defendants had a false positive rate of 32.42% and a false negative rate of 37.09%. There doesn't appear to be a meaningful difference between female and male defendants' false positive rates (a 0.99 ratio) or false negative rates (a 1.06 ratio), as the ratios of female to male were both close to 1.

Error rates in violent decile scores differed between groups: female defendants on the violent score had a lower false positive rate and higher false negative rate than male defendants. Female defendants had a false positive rate of 21.3% versus 29.7% for male defendants, and a false negative rate of 62.1% versus 44.9%.

Calibration – Cox Proportional Hazards Model and Survival Analysis

The calibration between female and male defendants was unequal. Out of all defendants classified as medium or high risk, 51% of female defendants reoffended compared to 64% of male defendants.

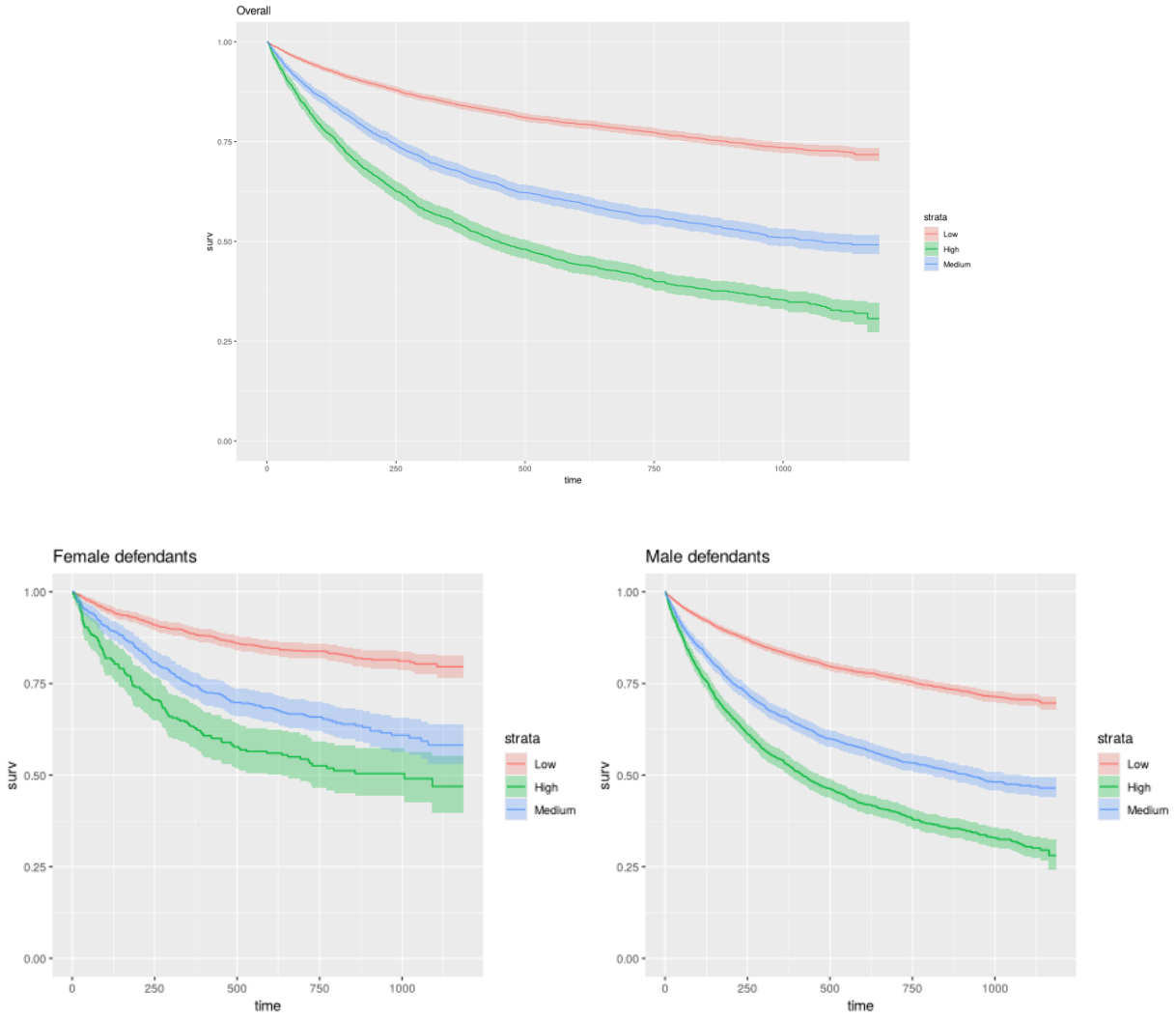


Figure 3. Share of defendants who did not reoffend over time by risk classification and sex: overall and by gender

A larger percentage of male defendants classified as high risk had reoffended by the end of the two-year period over which the survival analysis was conducted. At the end of the two years, 47.5% of high-risk female defendants had reoffended compared to 61.2% of male defendants. ProPublica's methodology reported these figures as well.

The Cox model with the interaction term found that female defendants reoffended less than male defendants with the same score. Female defendants' hazard of reoffending was about two-thirds that of male defendants across all risk categories–0.65 low risk, 0.69 medium risk, and 0.67 high risk–and had 95% confidence intervals below 1. The interaction terms were not significant, meaning that the gap between the hazard ratios was consistent across risk categories.

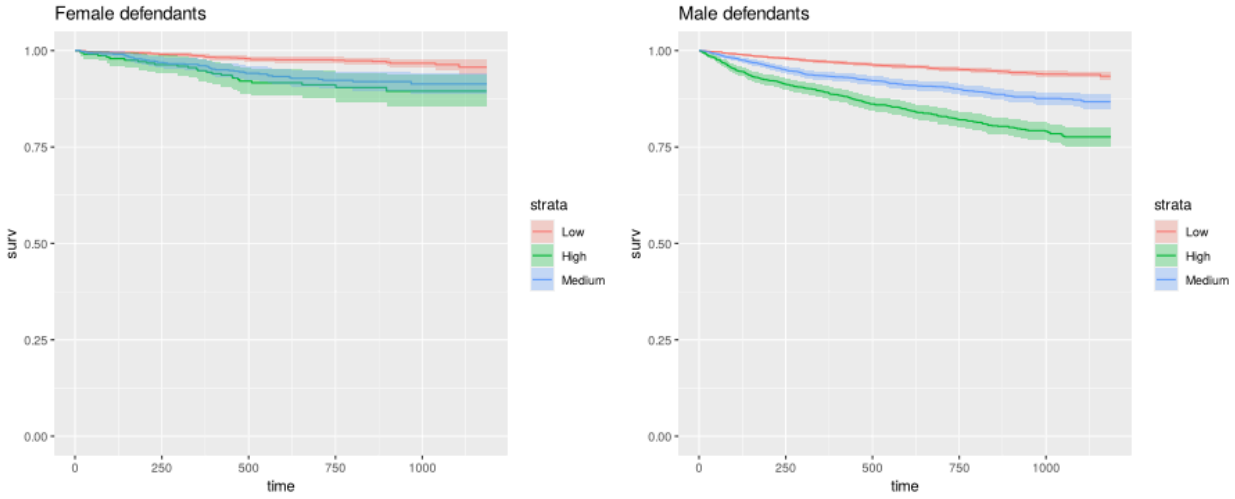


Figure 4. Share of defendants who did not reoffend violently over time, by COMPAS risk classification and sex

The violent score showed the same calibration pattern. In the Cox model with the interaction term, female defendants reoffended for violent offenses less often than male defendants with the same violent label: hazard ratios of 0.56 for low risk (95% CI 0.40–0.79), 0.70 for medium risk (0.50–0.97), and 0.48 for high risk (0.32–0.73).

Ranking

COMPAS ranked female and male defendants equally well within their own gender. When gender is treated as the protected class, the Cox model produced similar hazard ratios for female and male defendants when comparing each risk category with low-risk defendants of the same gender: for those rated high risk there was a hazard ratio of 3.50 for females and 3.40 for males, and for those rated medium risk there was a hazard ratio of 2.35 for females and 2.21 for males. Separate Cox models for each gender had concordance of about 0.63 for both, meaning COMPAS was a better predictor of recidivism risk than chance alone, but not by a substantial measure.

The ROC curves had similar AUCs for male and female defendants: females at 0.691 and males at 0.703. These are higher than the Cox concordance because they use the full 1–10

decile score rather than the three risk categories. The similar AUCs can be observed in the graphs: the curves for both groups are nearly on top of one another, indicating a similar ability to rank defendants by risk.

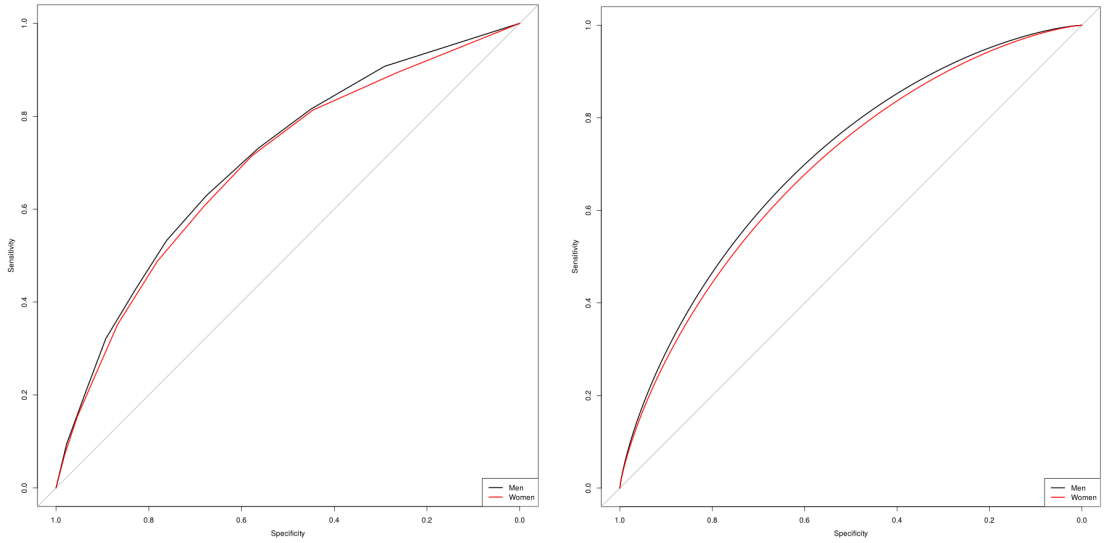


Figure 5. ROC curves by sex, unsmoothed and smoothed

Forest Plot

The forest plot's left panel indicates that the hazard ratio for female defendants compared with male defendants is similar across all three risk categories (0.65–0.69, all below 1). Female defendants labeled high-risk had the widest confidence interval, because fewer females were labeled high-risk. In the right panel, hazard ratios for high-risk and medium-risk defendants were similar between the sexes.

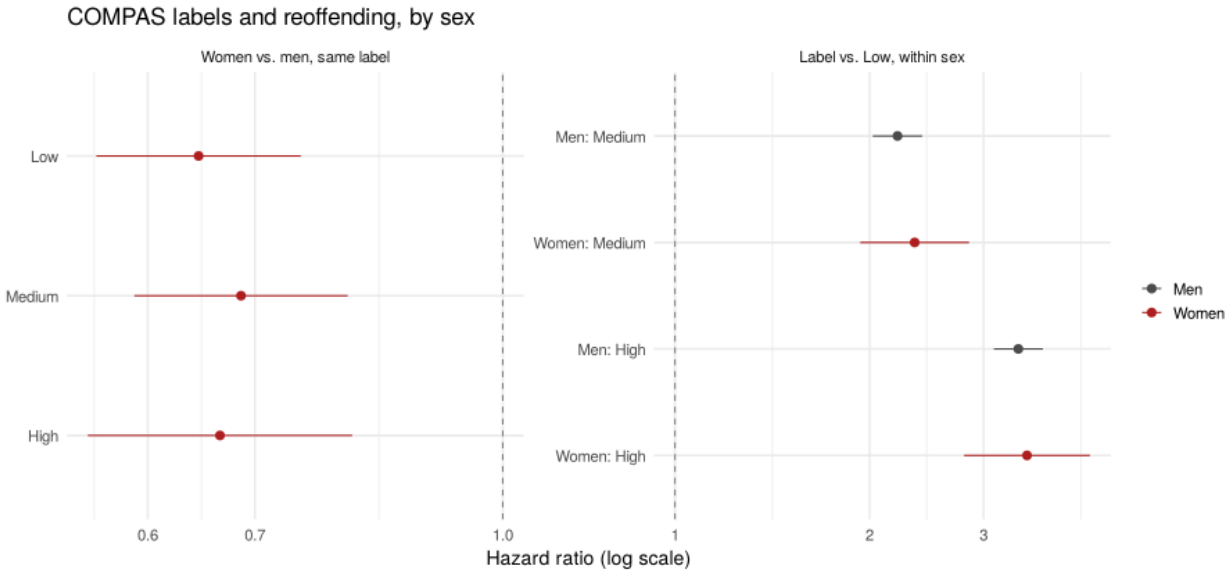


Figure 6. Forest plot of hazard ratios from the Cox model. Left: women vs. men. Right: medium and high vs. low within sex. Dots indicate hazard ratios, and the extending lines represent the 95% confidence intervals.

Limitations

Females are about one-fifth of the total sample, so estimates for female defendants have wider confidence intervals. This is especially true for female defendants classified as high-risk and for the violent score models, since few female defendants were labeled high-risk and only 105 recidivated violently.

The dataset categorizes gender as a binary: a person in the dataset is either female or male. As a result, there is a gap in measuring non-binary individuals, who may be incorrectly included in either gender category. We cannot know what share of the dataset these individuals make up and how much it impacts the results of the analyses.

Using data from only one county and a finite time frame limits the conclusions that can be drawn from this analysis. We do not know how the algorithm performs on groups with different demographic compositions or recidivism rates. As past research has found (Chouldechova 2017; Kleinberg et al. 2016), when two groups reoffend at different base rates, some form of bias will occur, as no score can satisfy both calibration and equal error rates. The bias from the COMPAS algorithm may be different for a different geographic location, or in the present, if changes to the algorithm have been made since this analysis (the data was from 2013–2014).

Conclusion

Based on these statistical tests, the COMPAS algorithm exhibits some bias against female defendants. While error rates were nearly the same between males and females, indicating that COMPAS made mistakes at similar rates for each gender, the same COMPAS score meant less risk for a female defendant than a male. Female defendants' hazard of reoffending was about two-thirds that of male defendants with the same score. Thus, the COMPAS algorithm overstates female defendants' risk. The violent score showed the same pattern: female defendants were less likely than comparable male defendants to receive a higher violent score, but those with the same violent score still reoffended for violent offenses less often than male defendants.

Since female and male defendants reoffended at different base rates (as was the case with race), it is impossible to satisfy both fairness tests at once. With regard to gender, the COMPAS algorithm sacrifices calibration in favor of equal error rates for the general scores, and the violent score meets neither test. This is the opposite of the bias measured for race, which was calibrated but failed on error rates. The real-world implication is that a judge may view a female defendant as higher risk than she truly is.